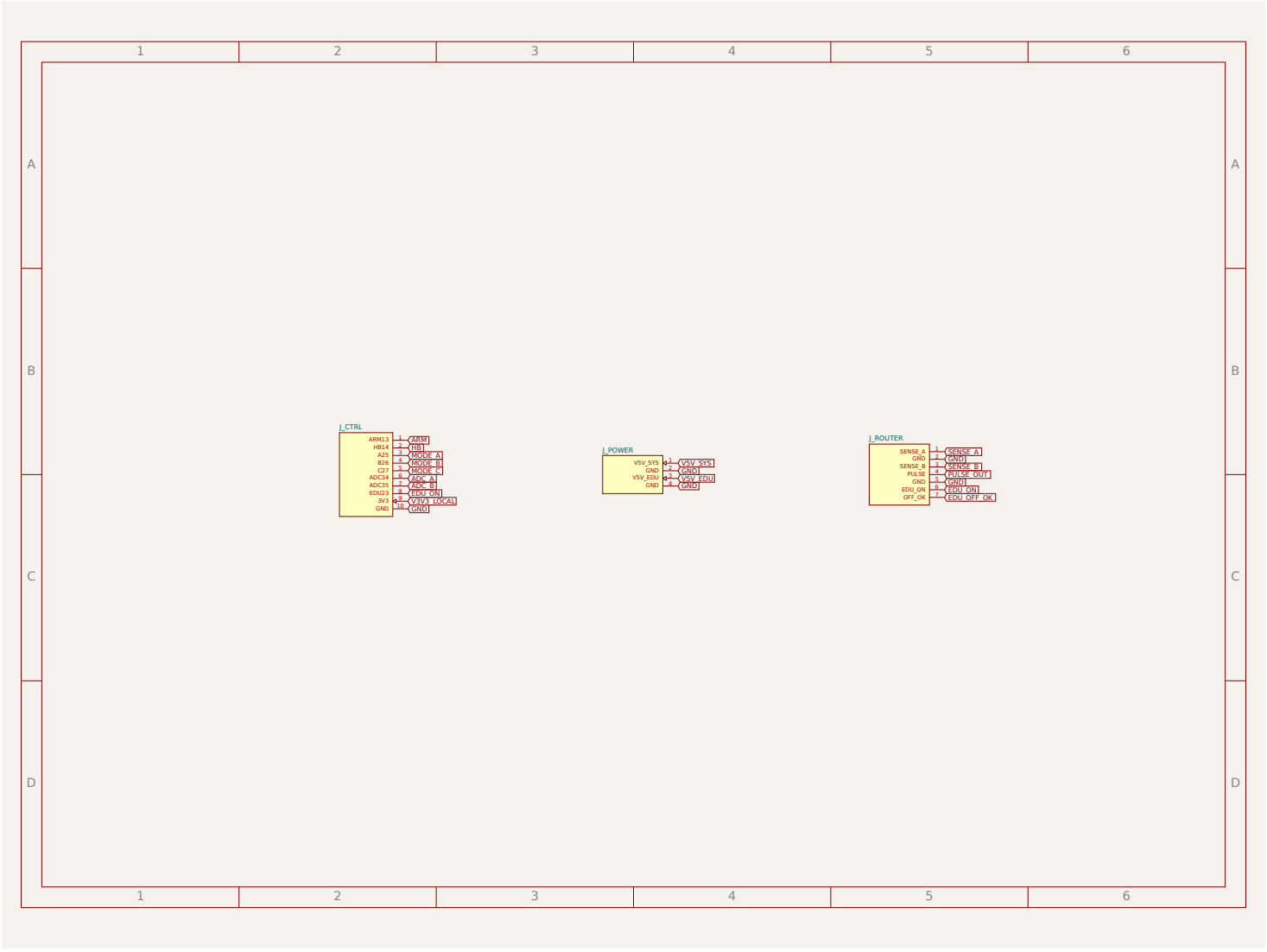
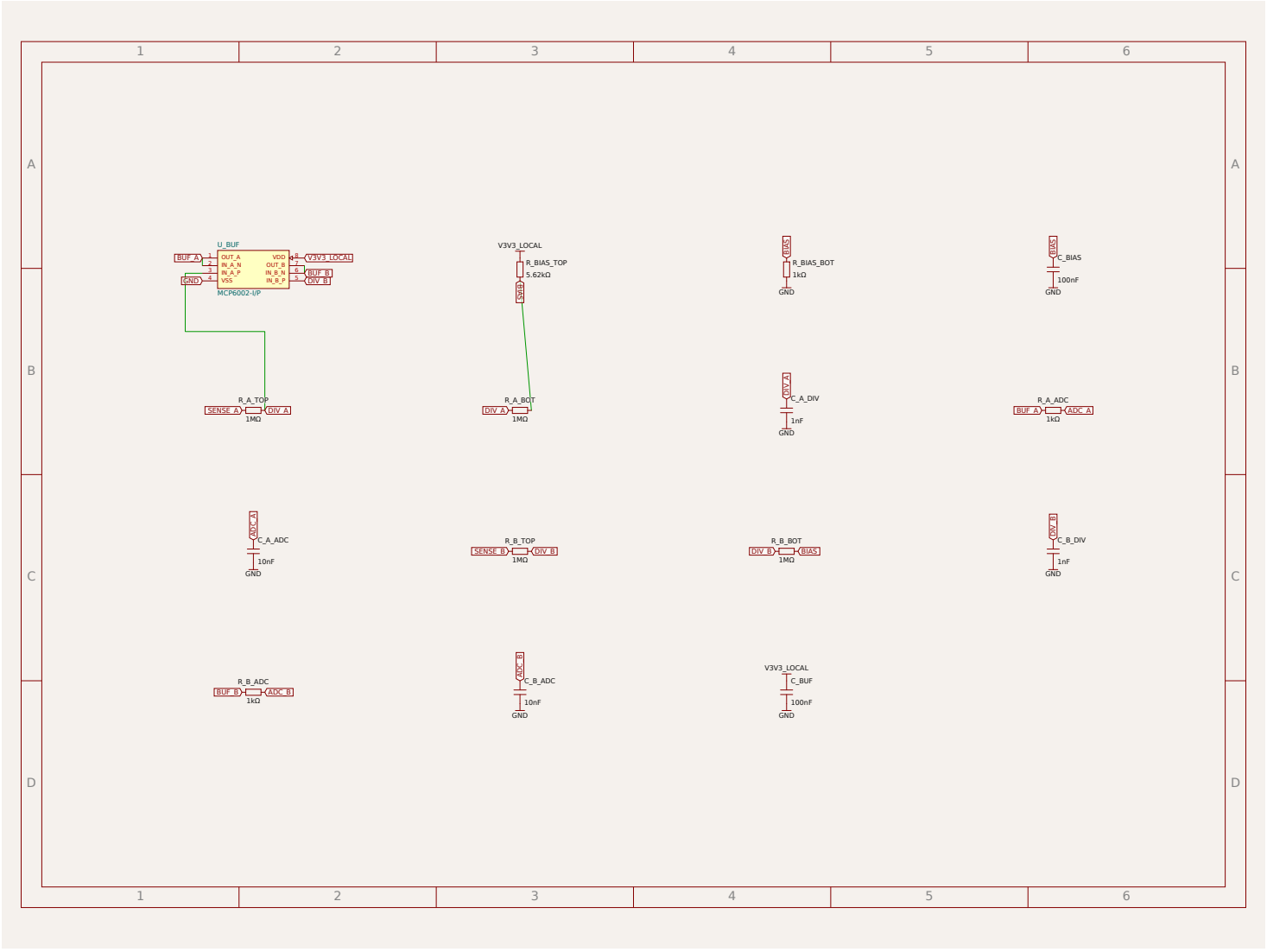
Placa inteligente — complemento elétrico

Fontes e ligações externas conforme guia de fiação. Matrizes de seleção não são parte deste esquema.

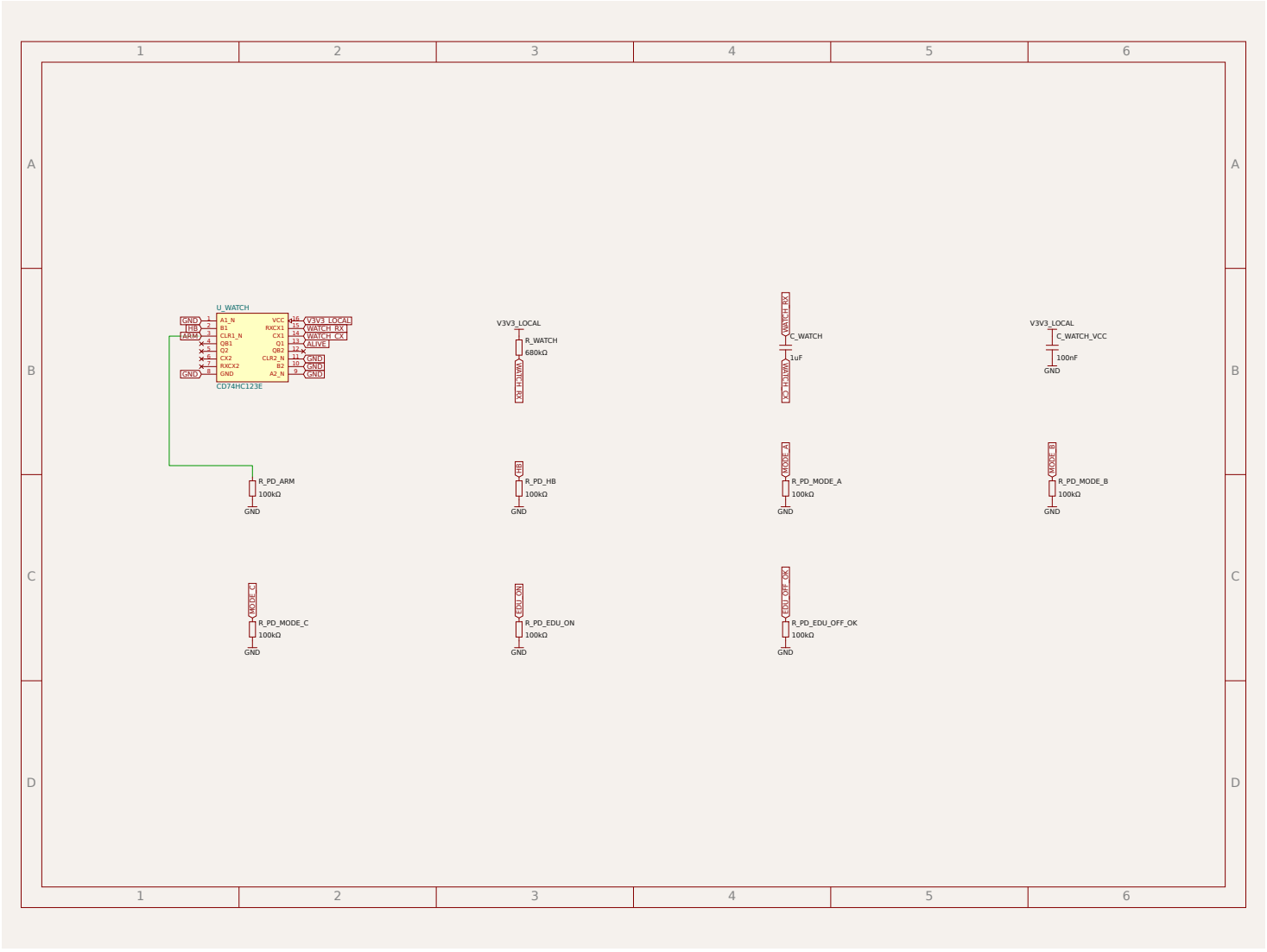
Interfaces


1
2
3
4
5
6
A
B
C
D
J_CTRL
ARM13
HB14
A25
B26
C27
ADC34
ADC35
EDU23
3V3
GND
ARM
HB
MODE_A
MODE_B
MODE_C
ADC_A
ADC_B
EDU_ON
V3V3_LOCAL
GND
J_POWER
V5V_SYS
GND
V5V_EDU
GND
V5V_SYS
GND
V5V_EDU
GND
J_ROUTER
SENSE_A
GND
SENSE_B
PULSE
GND
EDU_ON
OFF_OK
SENSE_A
GND
SENSE_B
PULSE_OUT
GND
EDU_ON
EDU_OFF_OK

Voltage

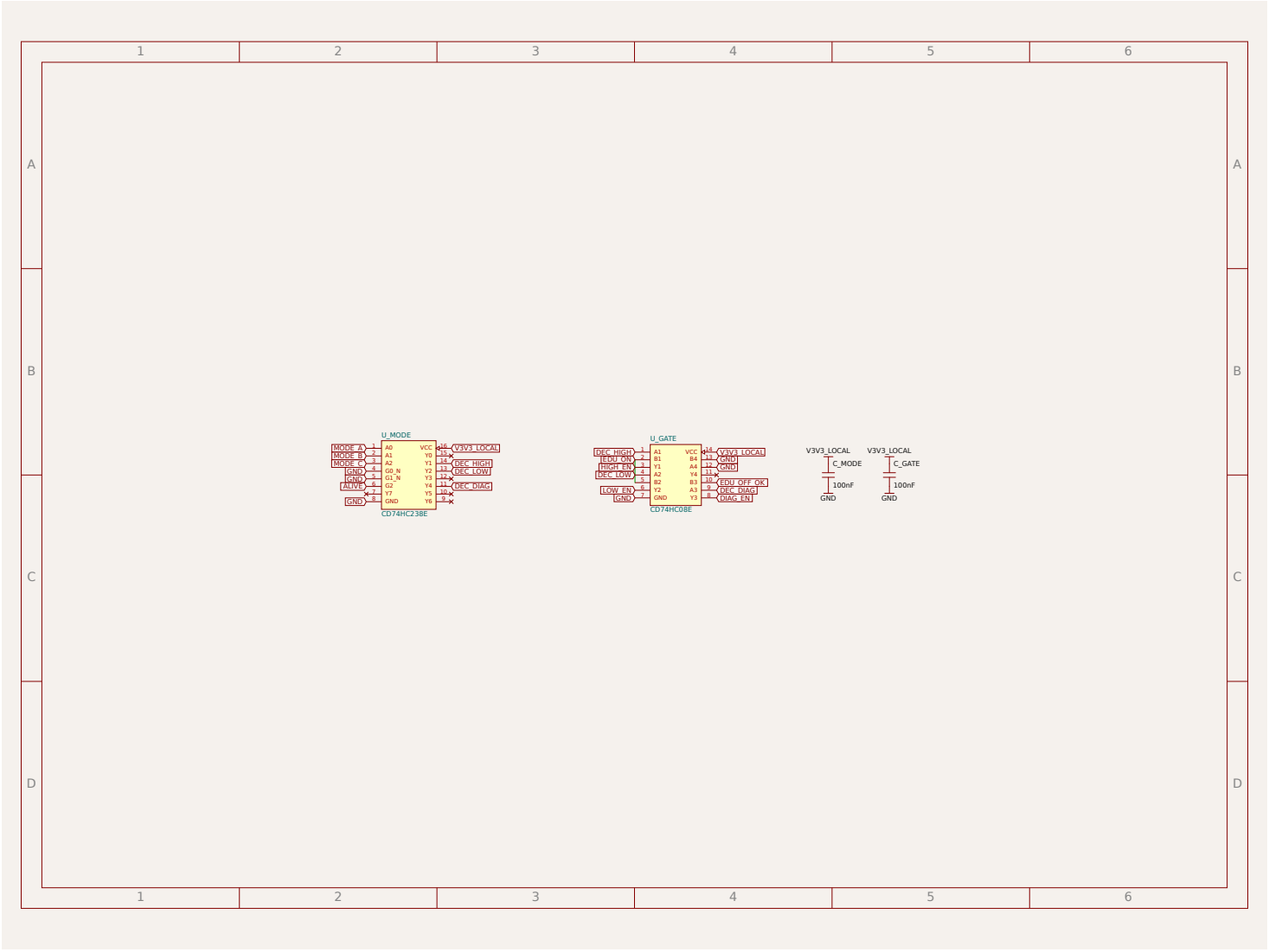

U_BUF
BUF_A
GND
OUT_A
IN_A_N
IN_A_P
VSS
VDD
OUT_B
IN_B_N
IN_B_P
V3V3_LOCAL
BUF_B
DIV_B
MCP6002-I/P
R_A_TOP
SENSE_A
DIV_A
1MΩ
V3V3_LOCAL
R_BIAS_TOP
5.62kΩ
R_A_BOT
R_BIAS_BOT
1kΩ
C_BIAS
100nF
C_A_DIV
1nF
R_A_ADC
ADC_A
1kΩ
C_A_ADC
10nF
R_B_TOP
SENSE_B
R_B_BOT
C_B_DIV
R_B_ADC
ADC_B
C_B_ADC
C_BUF

Watchdog

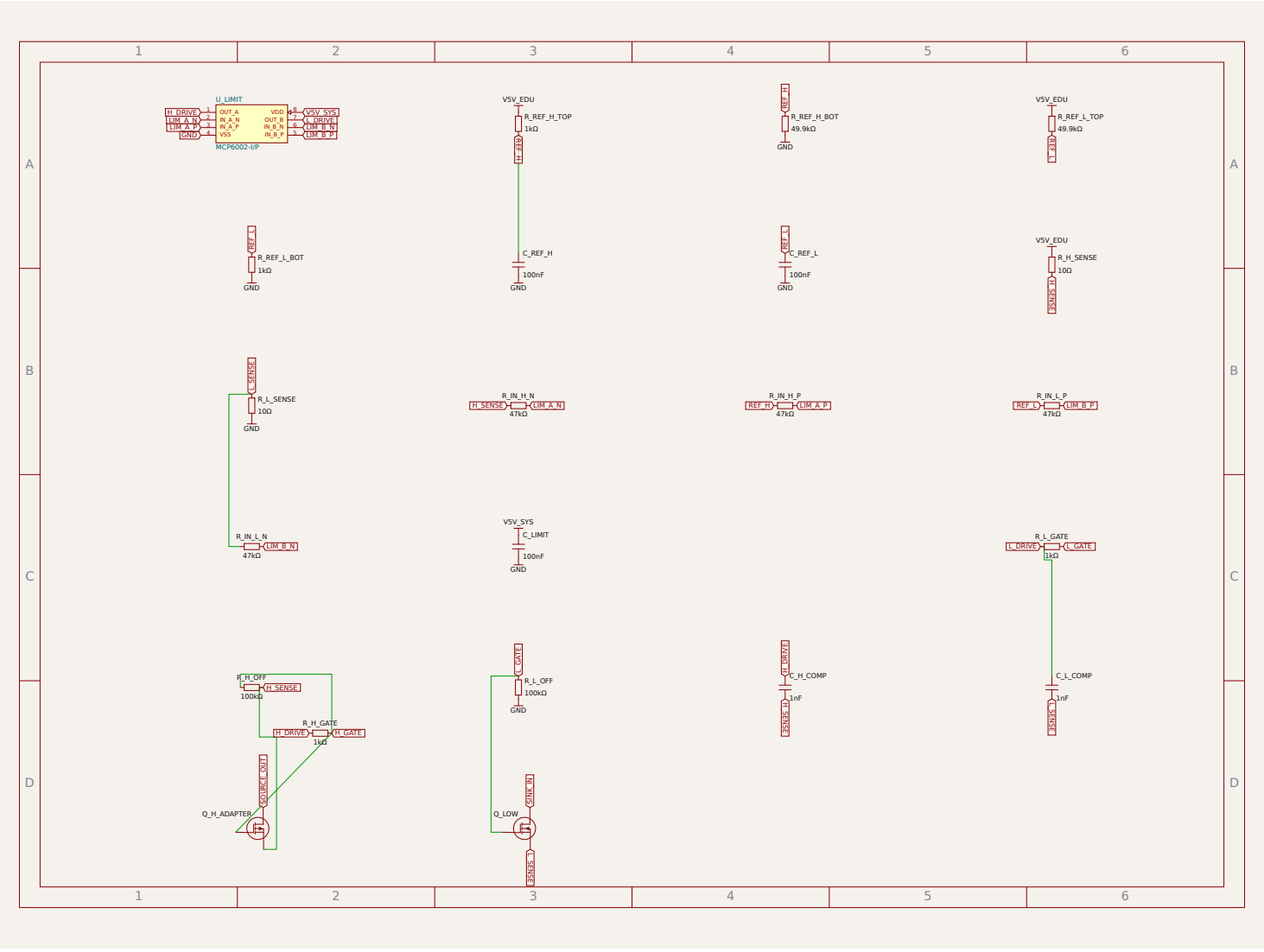


Placa inteligente · revisão final de projeto v1 · 4

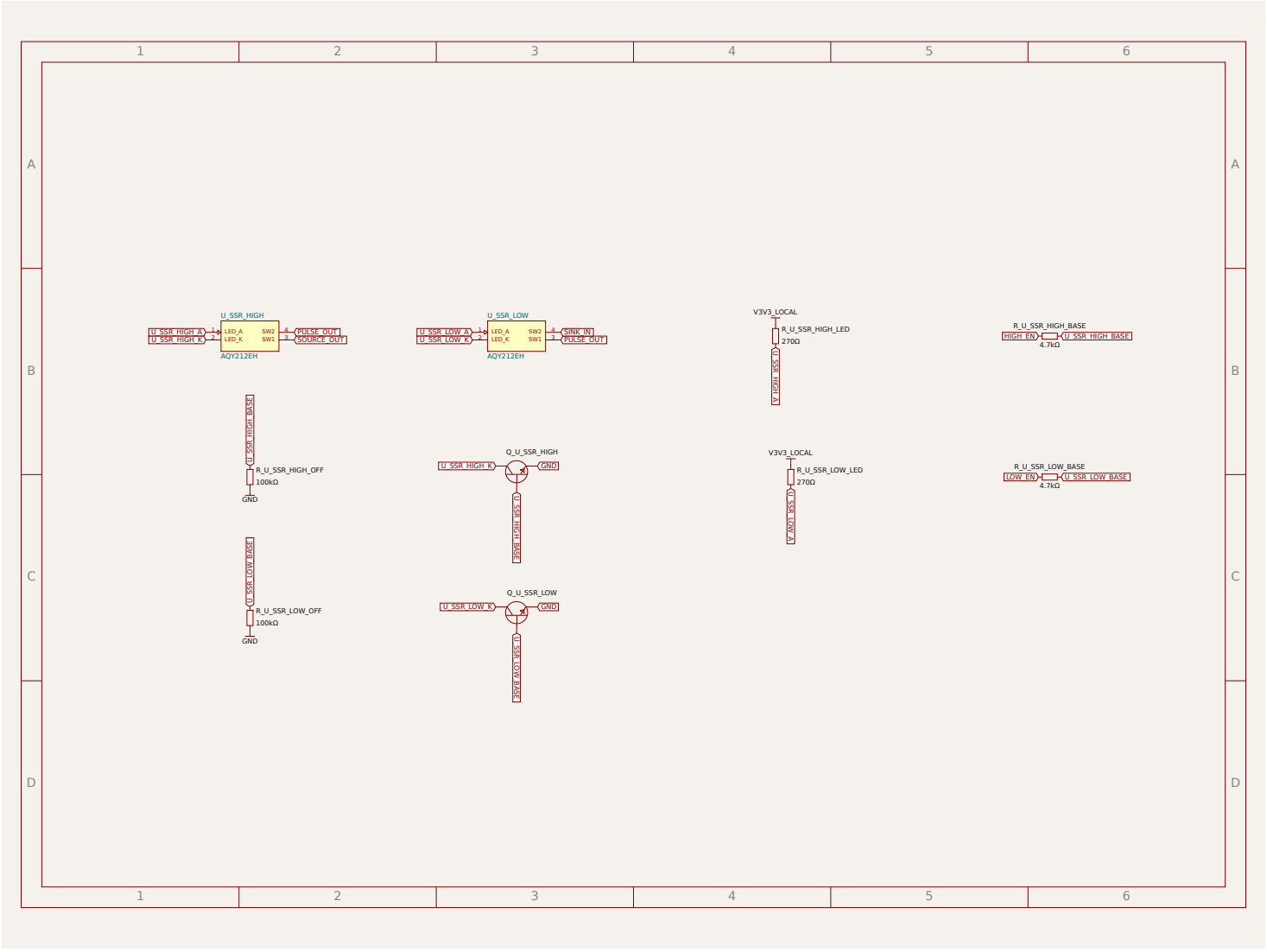
Mode


1
2
3
4
5
6
A
B
C
D
U_MODE
MODE_A
MODE_B
MODE_C
GND
GND
ALIVE
GND
A0
A1
A2
G0_N
G1_N
G2
Y7
GND
VCC
Y0
Y1
Y2
Y3
Y4
Y5
Y6
V3V3_LOCAL
DEC_HIGH
DEC_LOW
DEC_DIAG
CD74HC238E
U_GATE
DEC_HIGH
EDU_ON
HIGH_EN
DEC_LOW
LOW_EN
GND
A1
B1
Y1
A2
B2
Y2
GND
VCC
B4
A4
Y4
B3
A3
Y3
V2V3_LOCAL
GND
GND
EDU_OFF_OK
DEC_DIAG
DIAG_EN
CD74HC08E
V3V3_LOCAL
C_MODE
100nF
GND
V3V3_LOCAL
C_GATE
100nF
GND

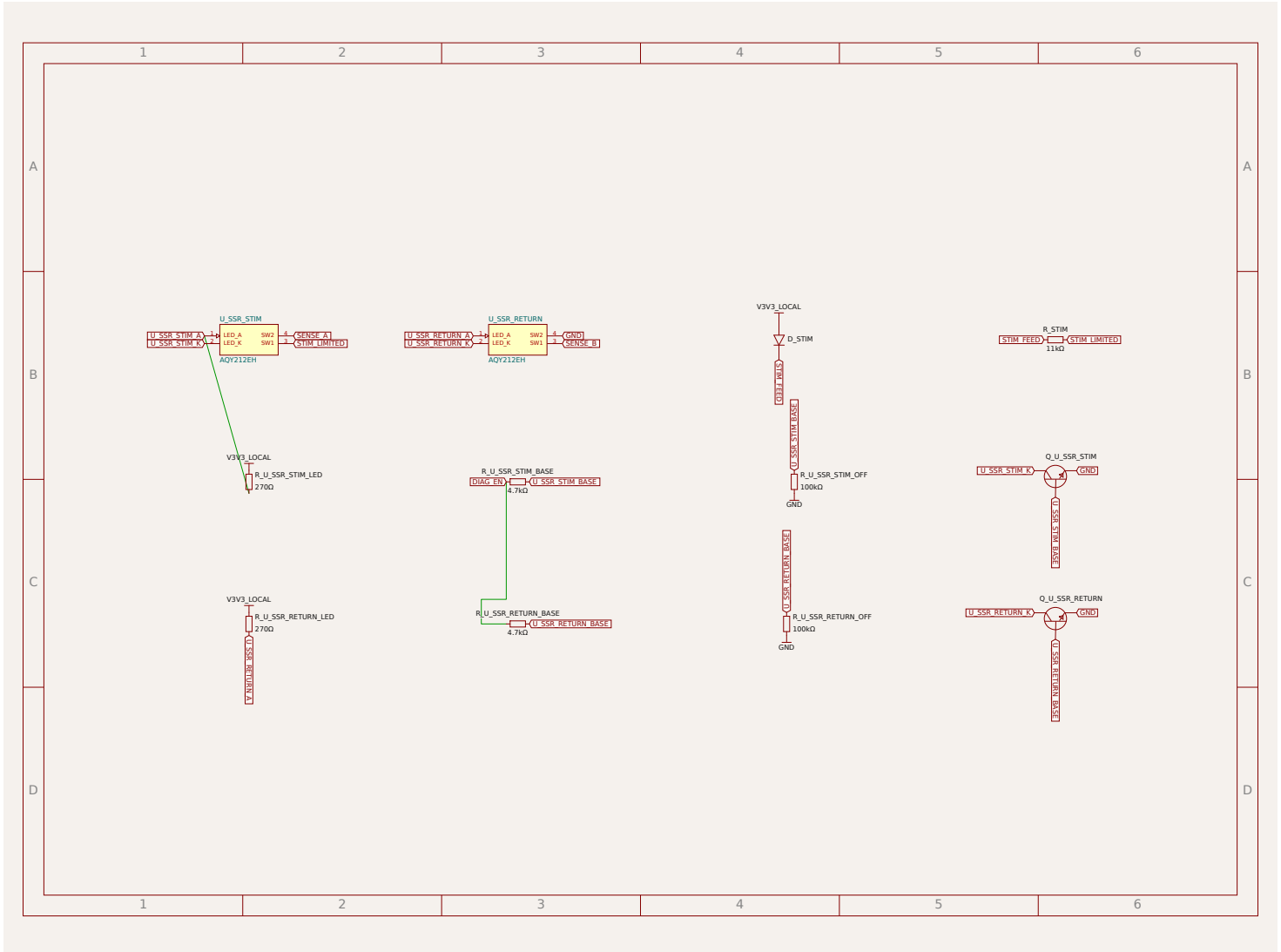
Limits


U_LIMIT
MCP6002-I/P
H_DRIVE
LIM_A_N
LIM_A_P
GND
VSS
OUT_A
IN_A_N
IN_A_P
VDD
OUT_B
IN_B_N
IN_B_P
V5V_SYS
L_DRIVE
LIM_B_N
LIM_B_P
V5V_EDU
R_REF_H_TOP
1kΩ
REF_H
C_REF_H
100nF
R_REF_H_BOT
49.9kΩ
R_REF_L_TOP
49.9kΩ
REF_L
R_REF_L_BOT
C_REF_L
R_H_SENSE
10Ω
SENSE_H
L_SENSE
R_L_SENSE
R_IN_H_N
H_SENSE
47kΩ
R_IN_H_P
R_IN_L_P
R_IN_L_N
C_LIMIT
R_L_GATE
L_GATE
C_H_COMP
C_L_COMP
1nF
R_H_OFF
100kΩ
R_H_GATE
H_GATE
SOURCE_OUT
Q_H_ADAPTER
R_L_OFF
GNX_IN
Q_LOW
SENSE_L

Pulse


U_SSR_HIGH
U_SSR_HIGH_A
U_SSR_HIGH_K
LED_A
LED_K
SW2
SW1
PULSE_OUT
SOURCE_OUT
AQY212EH
U_SSR_LOW
U_SSR_LOW_A
U_SSR_LOW_K
SINK_IN
PULSE_OUT
AQY212EH
V3V3_LOCAL
R_U_SSR_HIGH_LED
270Ω
U_SSR_HIGH_A
R_U_SSR_HIGH_BASE
HIGH_EN
4.7kΩ
U_SSR_HIGH_BASE
R_U_SSR_HIGH_OFF
100kΩ
GND
Q_U_SSR_HIGH
U_SSR_HIGH_K
GND
U_SSR_HIGH_BASE
V3V3_LOCAL
R_U_SSR_LOW_LED
270Ω
U_SSR_LOW_A
R_U_SSR_LOW_BASE
LOW_EN
4.7kΩ
U_SSR_LOW_BASE
R_U_SSR_LOW_OFF
100kΩ
GND
Q_U_SSR_LOW
U_SSR_LOW_K
GND
U_SSR_LOW_BASE

Diagnostic



Placa inteligente · revisão final de projeto v1 · 8